

Rohan Sada

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SUMMARY

AI/ML Software Engineer with 3+ years of production experience building scalable deep learning systems, computer vision pipelines, and distributed training frameworks. MS in ECE at Rutgers University (GPA 3.8/4.0) specializing in Machine Learning, Spatial AI, and Sensor Fusion. IEEE-published with expertise in model optimization, large-scale GPU training, and end-to-end ML pipeline development. Recipient of the AllGoVision Excellence Award.

TECHNICAL SKILLS

Core Machine Learning, Deep Learning, Computer Vision, Neural Networks, NLP, Model Optimization, Distributed Training, MLOps, Embedded AI, LangGraph, LangChain

Languages Python, C, C++, JavaScript, MATLAB

DL/Vision PyTorch, TensorFlow, HuggingFace, OpenCV, TensorRT, OpenVINO, FSDP, DeepSpeed, SLAM, LiDAR, ONNX

Tools Git, Docker, Linux, CI/CD, REST APIs, AWS/GCP, Kubernetes, NVIDIA Jetson, CUDA

WORK EXPERIENCE

AI Vision Engineer

Apr 2022 – Aug 2024

AllGoVision Technologies, Bengaluru, India

- Reduced model inference latency by **30%** and enabled production-scale real-time analytics across **20+** feeds by optimizing deep learning inference pipelines with **NVIDIA TensorRT** and a multi-threaded C++/Python architecture.
- Improved cross-camera person and vehicle **re-identification accuracy by 2x** by implementing **spatio-temporal matching algorithms** that track only high-probability targets, reducing system compute overhead.
- Architected a **camera calibration pipeline** using **geometric vision** to project 2D pixels into global EPSG 3D coordinates, significantly improving cross-camera object continuity in a production multi-camera tracking (MCT) system.
- Packaged the MCT pipeline into Docker executables for remote client-site deployment, managing the live setup, testing and debugging to ensure system reliability in production.

INTERNSHIPS

Research Intern – WINLAB

May 2025 – Aug 2025

WINLAB, Rutgers University, New Brunswick, NJ

- Architected an end-to-end streaming data pipeline integrating a spatial-temporal LSTM network to forecast urban mobility demands and optimized inference, serving real-time predictions across a live interactive map engine.
- Engineered an automated active-learning ingestion pipeline that processed 100+ hours of raw video; implemented a YOLO-driven pseudo-labeling framework that eliminated 85% of manual annotation bottlenecks and accelerated downstream model retraining cycles.

ACADEMIC PROJECTS

DataLens

- Built DataLens, a conversational analytics product that translates natural language to SQL using a GRPO-trained Qwen2.5-Coder-7B model with execution based rewards.
- Designed a 3-iteration self-correction loop that re-prompts the model with SQL error messages on failed queries.

3D Object Detection Extension for SLAM3R

- Developed a DETR-style transformer detection head integrated into SLAM3R for simultaneous 3D reconstruction and bounding box prediction from multi-view images.
- Trained only 8.4% of total parameters by leveraging frozen ViT (Vision Transformer) features from the I2P model.
- Managed the dataset pipeline of 13,000+ indoor multi-view images and 3D points.
- Achieved 51% mAP@25 on 3D object detection benchmarks validating end-to-end spatial perception accuracy.
- Demonstrates end-to-end 3D perception pipeline from raw images to spatially aware object detection which can be directly applicable to autonomous navigation.

Distributed BLIP-2 Fine-Tuning

- Fine-tuned BLIP-2 on medical data using PyTorch FSDP and DeepSpeed across 16 A100 GPUs reducing memory usage by 34% and achieving a 10.7x throughput speedup.
- Benchmarked DDP, FSDP and DeepSpeed ZeRO-3 across 4, 8, 16 A100 GPUs on the NERSC Purlmutter supercomputer, demonstrating super-linear scaling for shared strategies versus sub-linear scaling for DDP due to communication overhead.
- ZeRO-3 achieved 91% average GPU utilization with peak memory dropping from 28.5 GB to 12.4GB across 4 to 16 GPUs.

EDUCATION

MS, Electrical & Computer Engineering

Rutgers University, New Brunswick 2024-2026
GPA: 3.8/4.0

B.Tech, Electrical & Computer Engineering

PES University, Bengaluru Graduated 2022
CGPA: 8.11/10.0

PUBLICATIONS

R. Sada, D. T. M and R. Rajesh, "Deep Learning Based Real-Time Fusion of Day and Thermal Videos," *IEEE TENSYP 2022*, Mumbai, India, pp. 1–6.
DOI: [10.1109/TENSYP54529.2022.9864438](https://doi.org/10.1109/TENSYP54529.2022.9864438)